

Appendix B: RS232 Command Set

RS232 Parameters

Baud rate: 9 600 ... 230 400
Data bits: 8
Parity: no
Stop bits: 1

Handshake: DTR-CTS

Interface cable requirements: 9pin Sub-D wired 1:1

OPO Controller	PC
2 TXD	RXD 2
3 RXD	TXD 3
5 GND	GND 5
1 DCD	DCD 1
6 DSR	DSR 6
7 RTS	RTS 7
4 DTR	DTR 4
8 CTS	CTS 8

Software Commands

Commands from PC to OPO Controller (ASCII characters)

- SHO

Shutter open
- SHC

Shutter closed
- GH,

Get Humidity

response: 1 Byte

(1) 00hex .. FFhex relative humidity [%]
- GPI,

Get Power pumplaser

response: 2 Bytes

(1) 00hex .. FFhex high byte of IR power [arbitrary]

(2) 00hex .. FFhex low byte

GPV,

Get Power OPO (signal)

response: 2 Bytes

- (1) 00hex .. FFhex high byte of VIS power [arbitrary]
- (2) 00hex .. FFhex low byte

GTS,

Get Temperature OPO-crystal

response: 2 Bytes

- (1) 00hex .. FFhex high byte of SHG temperature
[degrees centigrade * 10]
- (2) 00hex .. FFhex low byte

STSnnnn,

Set OPO-crystal temperature

nnnn (ASCII) – SHG temperature set value
[degrees centigrade * 10]

GZ,

Get Piezo Voltage

response: 2 Bytes

Pnnnn,

Set Piezo Voltage nnnn = ASCII '0'..'4095'

SLYnnnn,

Set Lyot Filter

nnnn = ASCII '-127'..' +127' lyot filter Steps (back / forward)

response: 1 Byte 00hex - ok
01hex - end switch pos. side
FFhex - end switch neg. side

SBnnnnnn,

Set Baud Rate

nnnnnn= ASCII

9600
19200
38200
57600
115200
230400

Spectrometer Commands:

GNn,

Set Spec Gain

n = (ASCII) '1' - factor 1
'2' - factor 3
'3' - factor 10
'4' - factor 30
'5' - factor 100
'6' - factor 300

GG,	Get spectrometer Gain response:	1 Byte	01hex - factor 1 02hex - factor 3 03hex - factor 10 04hex - factor 30 05hex - factor 100 06hex - factor 300
LZnnnnn,	Set Zoom Window middle wavelength nnnnn (ASCII) – middle wavelength of zoom window [Angstrom]		
SZI,	Zoom In		
SZO,	Zoom Out		
Fn,	SpecWindow width	n = ASCII	'1' for 5 nm '2' for 10 nm '3' for 20 nm '4' for 50 nm '5' for 100 nm '6' for 200 nm '7' for 500 nm
SONnn,	Set Offset	nnn= ASCII '0'..'255'	Spectrometer offset for the active gain setting
Wnnnn,	Spectrum Values	nnnn= ASCII '256','512','1024'	for setting of the spectrum data set length
GC,	GETCALIBRATION response:	8 Hex values	00 00 00 00 hex .. FF FF FF FFhex (unsigned long)(K0*1E12) 00 00 00 00 hex .. FF FF FF FFhex (unsigned long)(ALPHA0*1E9)

GSV,

Get spectral Peak Values

response: 8 Bytes

- (1) 00hex .. FFhex high byte of peak wavelength [Ångström]
- (2) 00hex .. FFhex low byte of peak wavelength [Ångström]

- (3) 00hex .. FFhex high byte of FWHM [Ångström]
- (4) 00hex .. FFhex low byte of FWHM [Ångström]

If this return value is FFFFhex a spectrometer overflow was detected.

- (5) 00hex .. FFhex high byte of center wavelength [Ångström]
- (6) 00hex .. FFhex low byte of center wavelength [Ångström]
- (7) 00hex .. FFhex high byte of lock state
- (8) 00hex .. FFhex low byte of lock state

lock state: 1 – OPO locked
0 – OPO not locked

GSH,

Get Spectrum All

Response: 2X wavelength values +12 Hex values

00hex .. FFhex high byte 1st value of screen, only Bit 0..3 significant

00hex .. FFhex low byte 1st value of screen

00hex .. FFhex high byte 256th value of screen, only bit 0..3 significant

00hex .. FFhex low byte 256th value of screen

or 512th

or 1024th

00hex .. FFhex high byte 1. wavelength (left margin of screen)

00hex .. FFhex low byte 1. wavelength

....

00hex .. FFhex high byte 256th wavelength (right margin of screen)

00hex .. FFhex low byte 256th wavelength

00hex .. FFhex high byte Peak wavelength

00hex .. FFhex low byte Peak wavelength

00hex .. FFhex high byte Spectrum FWHM

00hex .. FFhex low byte Spectrum FWHM

00hex .. FFhex high byte spectrum central wavelength

00hex .. FFhex low byte spectrum central wavelength

00hex

00hex .. 01hex OPO LOCK STATUS0 for unlocked

1 for locked

GSF, Get Spectrum Fast
 Response: 2x wavelength points + 8 Hex values
 00hex .. FFhex high byte 1st value of screen, only Bit 0..3 significant
 00hex .. FFhex low byte 1st value of screen
 .
 .
 00hex .. FFhex high byte 256th value of screen, only bit 0..3 significant
 00hex .. FFhex low byte 256th value of screen
 or 512th
 or 1024th

 00hex .. FFhex high byte 1st wavelength (left margin of screen)
 00hex .. FFhex low byte 1st wavelength
 .
 00hex .. FFhex high byte 256th wavelength (right margin of screen)
 00hex .. FFhex low byte 256th wavelength

 00hex .. FFhex high byte Peak wavelength
 00hex .. FFhex low byte Peak wavelength

 00hex .. FFhex high byte Spectrum FWHM
 00hex .. FFhex low byte Spectrum FWHM

Correlator commands

CAG,	Correlator Auto Gain	
CAVn,	Correlator Average	n = ASCII '1' '2' '4' .. '64'
GA,	Get Correlator Average	
	Response:	01hex .. 40hex for 1..64
CFn,	Correlator Filter	n = ASCII '0' for off '1' for on
GF,	Get Correlator Filter	
	Response:	00hex for off 01hex for on
CGnnn,	Correlator Gain	nnn = ASCII '0'..'255'
CPn,	Correlator Preamp	n = ASCII '0' for off '1' for on